

ALEX MORGAN CHEN

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SUMMARY

Software Engineer with 5+ years of experience designing and deploying scalable backend systems and machine learning pipelines. Delivered solutions that reduced infrastructure costs by 42% and improved system throughput by 3x at Fortune 500 companies. Passionate about turning complex data into actionable products. Seeking a senior engineering role where I can lead high-impact initiatives and mentor junior engineers.

EXPERIENCE

Senior Software Engineer | Stripe Inc.

2022 - Present

- Architected a real-time fraud detection pipeline processing 2.4M transactions/day, reducing chargebacks by 38%.
- Led a team of 6 engineers to migrate 14 microservices to Kubernetes, cutting deployment time from 45 min to 4 min.
- Optimized PostgreSQL query performance, achieving a 67% reduction in p99 latency across 3 critical endpoints.
- Designed and shipped a self-serve analytics dashboard used by 1,200+ internal stakeholders monthly.

Software Engineer | Databricks

2020 - 2022

- Developed a distributed ETL framework in Python and Spark that processed 8TB of daily data 2x faster than the legacy system.
- Built and deployed 4 ML models (XGBoost, LSTM) achieving 91% accuracy on churn prediction, generating \$2.3M in retained revenue.
- Automated CI/CD pipelines using GitHub Actions and Docker, reducing release cycle from 2 weeks to 3 days.
- Mentored 3 junior engineers through weekly 1-on-1s and code reviews, all of whom were promoted within 18 months.

Software Engineering Intern | Microsoft (Azure AI)

2019 - 2020

- Implemented a REST API serving 500K daily requests with 99.98% uptime over a 6-month deployment window.
- Researched and integrated BERT-based NLP models, improving intent classification accuracy from 74% to 89%.
- Presented findings to a team of 40 engineers and received the Intern Impact Award (top 5% of 800 interns).

EDUCATION

Harvard University — M.S. Computer Science

2018 - 2020

GPA: 3.43 / 4.00 | Thesis: *Efficient Attention Mechanisms for Long-Range Sequence Modeling*

Boston University — B.S. Computer Science, Minor in Mathematics

2014 - 2018

GPA: 3.91 / 4.00 | Summa Cum Laude | Dean's List all 8 semesters

SKILLS

Languages: Python, Java, Go, TypeScript, SQL, Bash, Scala

Frameworks: FastAPI, Django, React, Node.js, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy

Infrastructure: AWS, GCP, Docker, Kubernetes, Terraform, PostgreSQL, MongoDB, Redis, Kafka

Tools: Git, GitHub Actions, Airflow, Spark, Prometheus, Grafana, Jira, Confluence

PROJECTS

OpenMetrics — Open-source observability toolkit / github.com/alexmcchen/openmetrics

- Built a lightweight Prometheus-compatible metrics library in Go with 1,400+ GitHub stars and 80+ contributors.
- Reduced memory footprint by 55% vs. comparable tools through a custom lock-free ring buffer implementation.

NeuroSched — ML-powered Kubernetes scheduler (Harvard thesis project)

- Designed a reinforcement learning agent that improved cluster utilization by 29% over the default Kubernetes scheduler.
- Published results at MLSys 2020; paper received 140+ citations within 12 months of publication.

LANGUAGES

English (Native) | **Mandarin Chinese** (Fluent) | **German** (Intermediate — B2)

CERTIFICATIONS

AWS Certified Solutions Architect – Professional (2023) | Google Professional ML Engineer (2022) | CKA: Certified Kubernetes Administrator (2021)